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Quiz # 3

- ① For all values of k , why do we only need to consider only values of k' that lie within the 1st Brillouin zone? Also explain why there are only standing waves at Brillouin zone boundary.

* Referring to text (Kittel):

- values of k that are outside 1st Brillouin zone basically reproduce ~~the lattice motions~~ the lattice motions of the values found within the limits of $\pm \pi/a \rightarrow$ 1st Brillouin Zone

* Why only standing waves at Brillouin zone boundary?

- at the boundaries we find standing wave $\rightarrow Ue^{isk_a}$

* we know that $sk_{\max} a = \pm \pi$

* plug into top exponential eq'n:

* $Ue^{\pm i\pi} = U(-1)^s = \text{Standing wave}$

* with opposite phases, alternate atoms oscillate

* $U_s = \pm 1$ depends on sign of s

* wave doesn't move left or right

* this is equivalent to Bragg reflection of x-rays

* when Bragg ~~Condition~~ Condition is satisfied

* traveling waves cannot form nor move in a lattice

* Continuing (i)

* A standing wave is essentially created through back and ~~forth~~ forth reflections of the x-rays $\rightarrow$ this would satisfy the Bragg condition

(2) Vibrations of a Square Lattice

(a) show that the E.O.M. is:

$$M \frac{\partial^2 u_{lm}}{\partial t^2} = C \left[(u_{l+1,m} + u_{l-1,m} - 2u_{lm}) + (u_{l,m+1} + u_{l,m-1} - 2u_{lm}) \right]$$

* Atoms, under the condition of small displacements, can be thought of as the lattice of atoms ~~being~~ interconnected ~~by~~ by springs

* Force on atom in the lattice:

$$F_n = C(u_{n+1} - u_n) + C(u_{n-1} - u_n)$$

* Newton's 2nd Law also addresses ~~f~~ f:

$$\cancel{F} f = ma \Rightarrow F_n = M \left(\frac{\partial^2 u_n}{\partial t^2} \right)$$

$$M \left(\frac{\partial^2 u_n}{\partial t^2} \right) = C(u_{n+1} - u_n) + C(u_{n-1} - u_n)$$

$$M \left(\frac{\partial^2 u_n}{\partial t^2} \right) = -C(2u_n - u_{n+1} - u_{n-1})$$

* Cleaner version

Continue to next page!

② * Continuing ①

cb) Assume solutions of the form:

$$u_{xm} = u(0) \exp[i(k_x a + m k_y a - \omega t)]$$

* Show that E.O.M. is satisfied if:

$$\omega^2 M = 2C(1 - \cos k_x a - \cos k_y a)$$

* The above eq'n is essentially in the form of:

$$u_n = A \exp[i(q x_n - \omega t)] \quad (*)$$

* $x_n = \text{equilib. position of } n^{\text{th}} \text{ atom}$

* This is a solution to the eq'n found in (a) after boundary conditions have been applied

$$x_n = a_n$$

* we plug in eq'n (*) into eq'n found in part (a)

$$M \frac{\partial^2 u_n}{\partial t^2} = -\omega^2 \exp(i q a)$$

$$M(-\omega^2) \exp(i q a) = -C(1 e^{i q a} - e^{i q(n+1)a} - e^{i q(n-1)a})$$

$$M \omega^2 = \text{[scribbled out]}$$

$$\text{[scribbled out]} = C(1 - e^{i q a} - e^{-i q a})$$

$$\text{[scribbled out]} = 2C(1 - \cos q a)$$

* Continue on Back!

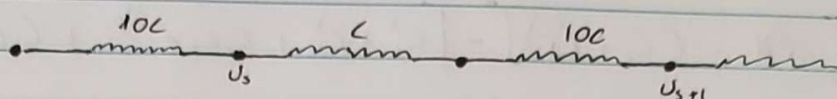
$$* M \omega^2 = \cancel{2C} 2C (1 - \cos qa)$$

$$* \omega^2 M = UC \sin^2 \frac{qa}{2}$$

$$* \text{Now } q_x = k_x \quad ; \quad q_y = k_y$$

$$* \text{So } \omega^2 M = 2C (1 - \cos k_x a - \cos k_y a)$$

③ Diatomic Chain



All Force Const's : $C = C_1$ and $10C = C_2$

• We need our E.O.M's first:

$$M\ddot{u}_{n,1} = C_1(u_{n,2} - u_{n,1}) - C_2(u_{n,1} - u_{(n-1),2})$$

$$\boxed{M\ddot{u}_{n,1} = C_1 u_{n,2} + C_2 u_{(n-1),2} - (C_1 + C_2) u_{n,1}}$$

$$M\ddot{u}_{n,2} = C_2(u_{n+1,1} - u_{n,2}) - C_1(u_{n,2} - u_{n,1})$$

$$\boxed{M\ddot{u}_{n,2} = C_2 u_{n+1,1} + C_1 u_{n,1} - (C_1 + C_2) u_{n,2}}$$

• Now we consider Normal Modes: Periodic Solutions

$$u_{n,1} = A_1 e^{i(kna - \omega t)} \quad ; \quad u_{n,2} = A_2 e^{i(kna - \omega t)}$$

• Plugging in $u_{n,1}$ and $u_{n,2}$ into above E.O.M's

$$A_1 \{ \omega^2 M - (C_1 + C_2) \} + A_2 (C_1 + C_2 e^{-ika}) = 0$$

$$A_1 (C_1 + C_2 e^{ika}) + A_2 \{ \omega^2 M - (C_1 + C_2) \} = 0$$

• Condition For Non-Trivial Solution

$$\det \begin{pmatrix} \omega^2 M - (C_1 + C_2) & C_1 + C_2 e^{-ika} \\ C_1 + C_2 e^{ika} & \omega^2 M - (C_1 + C_2) \end{pmatrix} = 0$$

• we know : $e^{ix} = \cos x + i \sin x$

$e^{-ix} = \cos x - i \sin x$

• Continue on back!

$$[\omega^2 M - (C_1 + C_2)] \times [\omega^2 M - (C_1 + C_2)] - (C_1 + C_2 e^{ika})(C_1 + C_2 e^{-ika}) = 0$$

$$[\omega^2 M - (C_1 + C_2)]^2 - [C_1^2 + C_2^2 + C_1 C_2 e^{-ika} + C_1 C_2 e^{ika}] = 0$$

$$[\omega^2 M - (C_1 + C_2)]^2 = [C_1^2 + C_2^2 + C_1 C_2 \{ \cos ka - i \sin ka \} + C_1 C_2 \{ \cos ka + i \sin ka \}]$$

$$[\omega^2 M - (C_1 + C_2)]^2 = [C_1^2 + C_2^2 + 2C_1 C_2 \cos ka]$$

$$\omega^2 M - (C_1 + C_2) = \sqrt{C_1^2 + C_2^2 + 2C_1 C_2 \cos ka}$$

$$\omega^2 M = C_1 + C_2 \pm \sqrt{C_1^2 + C_2^2 + 2C_1 C_2 \cos ka}$$

$$\omega^2 = \frac{1}{M} [C_1 + C_2 \pm \sqrt{C_1^2 + C_2^2 + 2C_1 C_2 \cos ka}]$$

Dispersion Relation

* Now $C_1 = C$ and $C_2 = 10C$

* So $\omega^2 = \frac{C}{M} (11 \pm \sqrt{101 + 20 \cos ka})$ @ $k=0$

$$= \sqrt{121 - 1} = 11$$

$$\omega^2 = \frac{C}{M} (11 + 11)$$

$$\omega = \sqrt{\frac{22C}{M}} \rightarrow \text{Optical}$$

$$\omega = 0 \rightarrow \text{Acoustic}$$

$$\omega^2 = \frac{C}{M} (11 \pm \sqrt{101 + 20 \cos \frac{\pi}{a} a})$$

$$\omega^2 = \frac{11C}{M} + \sqrt{101 + 20 + (-1)} = \frac{11C}{M} + \sqrt{81} = \frac{20C}{M}$$

$$\omega = \sqrt{\frac{20C}{M}} \rightarrow \text{Divide by 10 } \omega = \sqrt{\frac{2C}{M}}$$

$$\omega = \sqrt{\frac{20C}{M}} \rightarrow \text{Optical}$$

$$\omega = \sqrt{\frac{2C}{M}} \rightarrow \text{Acoustic}$$

@ $k = \frac{\pi}{a}$

